Machine Learning 101 for Financial Trading

FULL CODE EDITION

FSE-DAT-3019-1.1 | Data Analytics and Computational Modelling

Sector: Financial Services | Proficiency Level 3

LESSON PLAN

Tertiary Infotech Academy Pte Ltd

UEN 201200696W

Version 2.7 | 23 October 2026

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Version History

Version No	Date Effected	Change Update	Approved by
2.7	23 October 2026	Assessment papers rebuilt in the WSQ format for digital marking: cover page; page 2 with the trainee's name, NRIC, date and signature, the Instructions to Candidate and the Grading block with the assessor's name, NRIC, date and signature; questions from page 3, each with an answer box. Practical Performance redesigned around the labs, all in the Trading Simulator and TradingView: Task 1 hypothesis testing (Lab 4), Task 2 technical indicators and backtesting (Labs 5-7, 9), Task 3 machine-learning model and A/B testing (Labs 11, 13) and Task 4 the multi-agent desk with human approval, forward test, diagnosis and report (Labs 16, 18-20); marking points and the Assessor Key rewritten. Written Assessment Q6 (pairs trading) and Q9 (forms of data and turning headlines into data) now follow the slides.	Dr Alfred Ang
2.6	23 October 2026	Lab 17 (TradingView webhook) explained for learners: a new 'What This Lab Is For' section (what a webhook is, that the simulator supplies the URL, and the checks every alert passes before an order); Step 1 explains each part of the webhook card and alert message, Step 2 explains each of the seven self-test alerts and its HTTP code, and Step 3 gives the full real-alert walk-through (ngrok tunnel, webhook URL, chart, Pine strategy, alert dialog Condition / Message / Notifications, a quick Price-alert demo and where to watch the alert arrive). The Lab 17 screenshot now shows the TradingView webhook card with the 7/7 self-test result. Every lab in the deck is more visual: a 'Step by Step' slide (where to click, what to do, what you should see) and a 'How to Read the Screen' slide with numbered, colour-coded callouts on the simulator screenshot and a one-line takeaway; Lab 13 also gets a 'How to Reach the A/B Decision' slide (paired t-test, bootstrap Sharpe CI, Bonferroni alpha).	Dr Alfred Ang
2.5	23 October 2026	Topic 3 (K2): four slides added after 'Families of Learning Algorithms' explaining supervised, unsupervised and reinforcement learning - one slide each with a concept chart (SPY data for supervised and unsupervised, an agent-environment loop diagram for reinforcement), then a side-by-side comparison. Learner Guide section 3.1 and the Facilitator Guide Topic 3 teaching points cover the same content. Trading Simulator screenshots updated to the current app, whose Research tabs now follow the lab order: Strategy search and Walk-forward (Lab 10), ML models (Lab 11), Regimes (Lab 12), A/B test (Lab 13), Pairs screen (Lab 14), the multi-agent desk (Lab 20) and the 14-stage Pipeline Map (Getting Started); Lab 8 and Lab 9 screenshot captions aligned with the lab steps.	Dr Alfred Ang
2.4	22 October 2026	Practical Performance question paper simplified: one answer box per task, in which the candidate answers every part of the task and pastes its screenshot - the hypothesis test result in Task 1 and a TradingView Strategy Tester backtest of the Bollinger Bands strategy on XLF in Task 2.	Dr Alfred Ang

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2.3	22 October 2026	Assessment papers: every Practical Performance sub-question now has its own answer box, with screenshot boxes naming the lab where the step was practised, and the Written Assessment's ruled lines are replaced by answer boxes, so candidates type or paste their answers into the paper and upload it to the LMS. Assessment Plan Annex A wording aligned with the Written Assessment paper.	Dr Alfred Ang
2.2	22 October 2026	Screenshots added to Labs 7, 8 and 9: a Bollinger Band mean-reversion strategy in TradingView with the Paper Trading order panel (Lab 7), the simulator's paper order ticket and its passed pre-trade risk checks and fill (Lab 8), and Performance Analytics comparing a strategy with SPY buy-and-hold (Lab 9). Slide references in the Lesson Plan, Learner Guide and Facilitator Guide updated to the new slide numbers.	Dr Alfred Ang
2.1	21 October 2026	Low-code delivery. Every lab now runs in the Trading Simulator without coding, and the Practical Performance is done by configuring and running a provided trading program, with the same scenario, tasks, ability coverage and marking. A Prompt Library provides one AI prompt per lab, with one-click links, for learners who want code generated for them. Getting Started screenshots added (simulator Pipeline Map and Integrations, Alpaca paper dashboard, TradingView chart). Version History added to every document. Topic 2 slides added on mean reversion (the simulator's rules, mean reversion versus trend following, the main risk and safeguards), on the RSI look-back period (RSI(2) versus RSI(14)) and on long and short positions.	Dr Alfred Ang
2.0	18 October 2026	Major revision, focused entirely on quant and algorithmic stock trading. Courseware rebuilt and re-platformed to Alpaca (API-first broker, paper trading, 24/5 US stock trading) and TradingView (charting, Pine Script v6, Strategy Tester, Paper Trading) with Python. Four topics: quant success stories and the evolution of quant from Thorp and Renaissance through statistical arbitrage to high-frequency trading, the paradigm shift from machine learning to LLMs and its impact on quant, and AI agents versus agentic AI in trading; the classification of trading algorithms, indicators, strategies, a TradingView versus Alpaca comparison, order placement and metric definitions; machine learning models, statistical modelling, A/B testing and pairs trading as a hedging algorithm; and an end-to-end, automated algorithmic trading pipeline. 'How the algorithm works' slides explain each algorithm on real market data. Twenty hands-on labs build one pipeline stage by stage, each with an instruction PDF containing step-by-step AI prompts, and include live paper trading of a strategy. Assessment: Written Assessment (short-answer questions, K1-K9) and Practical Performance (four tasks, A1-A10) with a standalone SMART marking rubric. Facilitator Guide added. Fee restated in SGD exclusive of GST, optimum class size stated, provider name updated to Tertiary Infotech Academy Pte Ltd.	Dr Alfred Ang
1.1	22 December 2022	Multiple choice questions removed; practical test retained as the sole assessment instrument.	Dr Alfred Ang
1.0	10 June 2022	Initial release for pilot launch, aligned to Data Analytics and Computational Modelling (Proficiency Level 3).	Dr Alfred Ang

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1. Course Information

Item	Details
Course Title	Machine Learning 101 for Financial Trading
TSC Code and Title	FSE-DAT-3019-1.1 Data Analytics and Computational Modelling
Sector	Financial Services
TSC Category	Data Analytics and Information Technology Management
Proficiency Level	3
Course Duration	16 hours over 2 training days (14 hours classroom training + 2 hours assessment)
Daily Schedule	0900 - 1830 (8 instructional hours per day, excluding two 15-minute tea breaks and a 1-hour lunch break)
Mode of Delivery	Instructor-led classroom training with hands-on practical laboratory exercises on the Alpaca paper-trading platform, TradingView and Python
Software Platforms	Alpaca paper-trading account (Trading API and Market Data API, alpaca-py SDK); TradingView (charts, Pine Script v6, Strategy Tester, alerts); Python 3 with pandas, NumPy, scikit-learn, XGBoost, SciPy and matplotlib
Assessment Methods	Written Assessment - Short Answer Questions (45 minutes) | Practical Performance (75 minutes)
Training Venue	12 Woodlands Square, #07-85/86/87, Woods Square Tower 1, Singapore 737715
Class Size	Optimum class size of 20 participants (maximum 20)
Trainer to Learner Ratio	1 : 20
Course Fee	SGD 800 per participant (excluding GST)

2. Target Audience and Entry Requirements

Target Audience

- Traders, dealers, investment and research analysts in banks, brokerages and asset managers who want to test and automate trading ideas with data and machine learning.
- Quantitative, risk and data analysts in financial institutions who support trading, treasury or investment functions.
- Finance executives and technology professionals moving into algorithmic trading, trading technology or AI roles in financial services.
- Individual investors working in financial services who want a disciplined, evidence-based approach to systematic trading.

Entry Requirements

Knowledge and Skills

- Able to read, write, speak and listen in English at a proficiency level equivalent to the Employability Skills Workplace Literacy Level 5.
- Able to manipulate numbers at a proficiency level equivalent to the Employability Skills Workplace Numeracy Level 5.

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- Basic computer literacy; no programming is required - every lab runs in a low-code trading simulator.
- Basic understanding of financial markets (stocks, ETFs, buying and selling).

Experience

- Preferably 1 year of working experience in financial services or a related role involving data, investment or trading.

Equipment

- A laptop (macOS, Windows or Linux) with a modern web browser and internet access, able to run the free course Trading Simulator (installation is guided in Lab 1).
- A free Alpaca paper-trading account and a free TradingView account, created in class; no real money is used at any point.

3. Learning Outcomes

On completion of this course, participants will be able to:

LO	Learning Outcome	TSC Reference
LO1	Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.	A1, A6, K1, K9
LO2	Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.	A2, A3, K3, K5, K7
LO3	Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.	A4, A5, K2, K4, K6
LO4	Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.	A7, A8, A9, A10, K7, K8

4. TSC Alignment

This course is fully aligned to FSE-DAT-3019-1.1 Data Analytics and Computational Modelling at Proficiency Level 3: "Identify and utilise appropriate statistical algorithms and data models to perform data analyses, test hypotheses and derive inferences, patterns or solutions from data". All 10 ability statements and all 9 knowledge statements of the TSC are delivered and assessed.

Abilities

Ref	Ability Statement	Delivered In	Assessed By
A1	Develop hypotheses for implementing data analytics and computational modelling activities	Topic 1	PP Task 1
A2	Identify appropriate statistical algorithms and data models to test hypotheses and/or theories	Topics 2, 3	PP Task 2

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Ref	Ability Statement	Delivered In	Assessed By
A3	Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements	Topics 1, 2	PP Task 2
A4	Conduct statistical modelling of data to derive patterns and solutions	Topic 3	PP Task 3
A5	Perform A/B testing based on different hypotheses	Topic 3	PP Task 3
A6	Identify possible big data applications	Topic 1	PP Task 1
A7	Perform coding and configuration of software agents or programs based on selected models or algorithms	Topic 4	PP Task 4
A8	Conduct tests on the actions taken and outcomes to assess effectiveness of the models	Topic 4	PP Task 4
A9	Diagnose unintended outcomes produced by statistical models	Topic 4	PP Task 4
A10	Draw relevant trends and insights from data analysis to support decisions and report accordingly	Topic 4	PP Task 4

Knowledge

Ref	Knowledge Statement	Delivered In	Assessed By
K1	Data analytics and modelling business use cases	Topics 1, 4	WA Q1
K2	Types of statistical analyses, data models, algorithms and advanced computational methods	Topic 3	WA Q2
K3	Range and application of various statistical methods and algorithms	Topic 2	WA Q3
K4	Range and application of various types of data models	Topic 3	WA Q4
K5	Usage of analytics platforms and tools	Topics 1, 2, 4	WA Q5
K6	Statistical modelling techniques	Topic 3	WA Q6
K7	Coding languages for programming of algorithms and signals	Topics 2, 4	WA Q7
K8	Potential reasons for unintended outcomes	Topic 4	WA Q8
K9	Forms of data within financial services	Topic 1	WA Q9

5. Course Structure and Duration by Topic

Topic	Title	Day	Labs	TSC Ref	Duration	Slides
Topic 1	Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents	Day 1	Labs 1-4	A1, A6, K1, K9	3 hours	35 - 75
Topic 2	Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca	Day 1	Labs 5-9	A2, A3, K3, K5, K7	4.25 hours	76 - 140
Topic 3	Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals	Day 2	Labs 10-14	A4, A5, K2, K4, K6	2.5 hours	141 - 174
Topic 4	Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis	Day 2	Labs 15-20	A7, A8, A9, A10, K7, K8	3.25 hours	175 - 198
Admin	Course overview, ground rules, daily recaps	1, 2	-	-	1 hour	1 - 34
Assessment	Written Assessment and Practical Performance	2	-	K1-K9, A1-A10	2 hours	-
TOTAL					16 hours	

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Topic hours include the trainer-led content and the hands-on labs for that topic. Training 14 hours + assessment 2 hours = 16 hours, excluding tea and lunch breaks.

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Hands-on labs

Lab	Title	Topic	TSC Ref	Duration	Slide
1	Pipeline Skeleton, Configuration and the Alpaca Paper Account	1	K5	20 min	67
2	Data-Validation Stage: Cross-Check Pipeline Bars against TradingView	1	A3, K5, K9	20 min	68
3	Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality	1	A6, K9	20 min	70
4	Hypothesis Stage: from a Written Card to an Automated Statistical Test	1	A1, A6, K1	15 min	73
5	Indicator Stage: Pine Script v6 Indicators and their Python Twins	2	A3, K3, K7	25 min	123
6	Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation	2	A2, A3, K3, K7	30 min	126
7	Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals	2	A2, K3, K7	25 min	130
8	Execution Stage: an Order Manager that Trades the Paper Account in Any Session	2	A3, K5	30 min	133
9	Evaluation Stage: Performance and Risk Metrics with a Benchmark Table	2	A3, K3	25 min	136
10	Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay	3	A2, A4, K2	20 min	163
11	ML-Signal Stage: a Walk-Forward Classifier from Probability to Position	3	A4, K4, K6	20 min	166
12	Regime-Filter Stage: PCA and K-Means as a Gate on the Signals	3	A4, K4, K6	15 min	168
13	A/B-Test Stage: a Statistical Decision before Capital Moves	3	A5, K2	15 min	170
14	Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve	3	A4, K6	20 min	172
15	Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even	4	A7, A8, K7	25 min	187
16	Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading	4	A7, A8, K7	25 min	189
17	Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution	4	A7, K5, K7	20 min	191
18	Diagnostics Stage: Automated Bias Checks before Go-Live	4	A9, K8	20 min	192
19	Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report	4	A8, A10	25 min	194
20	Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval	4	A7, A10, K1	20 min	196

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6. Daily Training Schedule

Each training day comprises 8 instructional hours from 0900 to 1830, excluding two 15-minute tea breaks and a 1-hour lunch break. The 2-hour final assessment (45-minute Written Assessment and 75-minute Practical Performance) is conducted at the end of Day 2.

Day 1

Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
09:00 - 09:30	Course Overview, Trainer Introduction and Ground Rules Course objectives, TSC alignment, assessment overview, ground rules and an icebreaker on how participants use data in trading or investment today.	-	Trainer presentation; Slides; Learner Guide	-
09:30 - 10:45	T1.1-1.4 Quant Success Stories, the Evolution of Quant, the ML-to-LLM Shift, and Agentic AI versus AI Agents Quant success stories (Thorp, Renaissance); the evolution of quant to high-frequency trading and the six paradigm shifts; machine learning to LLMs and the impact on quant; agentic AI versus AI agents and multi-agent trading systems.	LO1	Trainer presentation, discussion and worked examples; Slides; Learner Guide	K1, K9
10:45 - 11:00	Tea Break			
11:00 - 11:30	T1.5-1.6 Forms of Financial Data, Big Data Applications and Trading Hypotheses Market, reference, fundamental and alternative data; the four Vs; big data applications; writing a testable hypothesis.	LO1	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A1, A6, K1, K9
11:30 - 11:50	Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account Create the pipeline package and its config.yaml (universe, strategy parameters, risk limits, costs, secret names), build a validating configuration loader and a market-clock module, and run `python -m pipeline check` against the Alpaca paper account or the offline simulator.	LO1	Hands-on lab; Learner Guide; lab folder Lab01-alpaca-setup	K5
11:50 - 12:10	Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView Read daily bars for SGX:D05 and NASDAQ:AAPL from the TradingView Data Window, record them, and build a validation stage that compares TradingView with the pipeline's raw and dividend-adjusted bars and writes a validation report.	LO1	Hands-on lab; Learner Guide; lab folder Lab02-tradingview-charts	A3, K5, K9
12:10 - 12:30	Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality Build `python -m pipeline ingest` to fetch and cache daily, weekly, monthly and hourly bars for the universe, run automated data-quality checks, ingest headlines as unstructured data, and describe the forms of data and the 3 Vs of big data.	LO1	Hands-on lab; Learner Guide; lab folder Lab03-alpaca-market-data	A6, K9
12:30 - 13:30	Lunch Break			
13:30 - 13:45	Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test Write a falsifiable hypothesis card in YAML (H1, H0, signal known at the close, outcome, test, decision rule, big-data use case) and build `python -m pipeline hypothesis` to test it with a Welch t-test, a bootstrap confidence interval and a robustness grid.	LO1	Hands-on lab; Learner Guide; lab folder Lab04-trading-hypothesis	A1, A6, K1

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
13:45 - 14:30	T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies SMA, EMA, RSI, MACD, Bollinger Bands, ATR, VWAP, OBV; trend following, mean reversion, momentum, breakout, pairs trading and execution algorithms.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A2, K3
14:30 - 14:55	Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins Add the desk's Pine Script v6 indicators to a TradingView chart, build pipeline/indicators.py with twins of ta.ema, ta.sma, ta.rsi, ta.bb, ta.atr, ta.macd, ta.crossover and ta.crossunder, and cross-check the values against the TradingView Data Window.	LO2	Hands-on lab; Learner Guide; lab folder Lab05-pine-indicators	A3, K3, K7
14:55 - 15:25	Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation Configure a Pine Script v6 strategy with costs, read the Strategy Tester, build `python -m pipeline reconcile` to check the Pine settings and reproduce the results, and place and manage paper trades from the chart with TradingView Paper Trading (and optionally an Alpaca paper account).	LO2	Hands-on lab; Learner Guide; lab folder Lab06-pine-strategy-tester	A2, A3, K3, K7
15:25 - 15:40	Tea Break			
15:40 - 16:05	Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals Build pipeline/signals.py with trend, mean-reversion, momentum and breakout signals that return target positions with next-bar alignment, a registry, an automated look-ahead test and a universe target file for the execution stage.	LO2	Hands-on lab; Learner Guide; lab folder Lab07-technical-indicators	A2, K3, K7
16:05 - 16:50	T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement TradingView, Alpaca Trading and Market Data APIs, Python libraries; Pine Script and Python; order types, time in force, bracket orders and the order lifecycle.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A3, K5, K7
16:50 - 17:20	Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session Build pipeline/execution.py: a session-aware, idempotent order manager that sizes orders from target weights, sends market, extended-hours limit, GTC-queued or bracket PAPER orders, polls their lifecycle, cancels protective legs before exits, and falls back safely when an order is rejected.	LO2	Hands-on lab; Learner Guide; lab folder Lab08-paper-orders	A3, K5
17:20 - 17:50	T2.5 Performance and Risk Metric Definitions CAGR, volatility, Sharpe, Sortino, Calmar, maximum drawdown, win rate, profit factor, expectancy, exposure, turnover, alpha and beta.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A3, K3
17:50 - 18:15	Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table Build `python -m pipeline evaluate` to compute total return, CAGR, volatility, Sharpe, Sortino, maximum drawdown, Calmar, win rate, profit factor, exposure, turnover, alpha and beta for every signal against a buy-and-hold benchmark, and interpret the ranking.	LO2	Hands-on lab; Learner Guide; lab folder Lab09-performance-metrics	A3, K3
18:15 - 18:30	Day 1 Recap and Q&A Consolidation of Topics 1 and 2; preview of Day 2.	-	Trainer presentation; Slides; Learner Guide	-

Total instructional time, Day 1: 8 hours (480 minutes), excluding the tea breaks and the 1-hour lunch break.

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Day 2

Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
09:00 - 09:15	Day 1 Recap and Day 2 Overview Review of Topics 1 and 2; orientation to Topics 3 and 4 and to the assessment.	-	Trainer presentation; Slides; Learner Guide	-
09:15 - 09:40	T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models Descriptive to prescriptive analysis; supervised, unsupervised and reinforcement learning; strategy search; regression, trees, random forest, XGBoost, neural networks.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A4, K2, K4
09:40 - 10:00	Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay Build `python -m pipeline search` to grid-search three signal families, rank parameter sets in sample, measure out-of-sample decay, prefer a robust parameter region, and decide whether anything deserves promotion into config.yaml.	LO3	Hands-on lab; Learner Guide; lab folder Lab10-strategy-grid-search	A2, A4, K2
10:00 - 10:20	Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position Build pipeline/ml.py: engineer causal features, train logistic regression, random forest and XGBoost with walk-forward TimeSeriesSplit validation, turn P(up) into a target position and backtest it after costs.	LO3	Hands-on lab; Learner Guide; lab folder Lab11-ml-signal-classifier	A4, K4, K6
10:20 - 10:35	Tea Break			
10:35 - 10:55	T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques K-means, hierarchical clustering and PCA for regimes; returns and stationarity; time-ordered validation; feature engineering; regularisation.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A4, K4, K6
10:55 - 11:10	Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals Build pipeline/regime.py: PCA on the universe's returns, K-means regimes on backward-looking SPY features, walk-forward regime labels, and a gate that forces signals flat in the stress regime, evaluated after costs.	LO3	Hands-on lab; Learner Guide; lab folder Lab12-market-regimes	A4, K4, K6
11:10 - 11:25	T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm Hypotheses, Welch's t-test, bootstrap Sharpe difference, multiple testing; cointegration, hedge ratio, spread z-score and market neutrality.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A4, A5, K2, K6
11:25 - 11:40	Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves Build `python -m pipeline abtest` to compare any two library signals with a Welch t-test, a paired t-test and a block-bootstrap confidence interval for the Sharpe difference, and to record an accept or reject decision with a multiple-testing caution.	LO3	Hands-on lab; Learner Guide; lab folder Lab13-ab-testing	A5, K2
11:40 - 12:00	Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve Build pipeline/pairs.py: screen pairs with the Engle-Granger cointegration test, estimate the hedge ratio, trade the spread z-score out of sample with next-bar execution and costs on both legs, confirm market neutrality and output leg targets.	LO3	Hands-on lab; Learner Guide; lab folder Lab14-pairs-trading	A4, K6
12:00 - 12:30	T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs The eight-stage pipeline; configuration and pre-trade risk controls; next-bar execution; commission, spread, slippage and regulatory fees.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A7, A8, K7
12:30 - 13:30	Lunch Break			

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
13:30 - 13:55	Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even Build pipeline/backtest.py around the common backtester: run any library signal under every cost scenario (and a custom one), break the cost into commission, spread, slippage and regulatory fees, find each strategy's break-even cost and test next-open versus next-close execution.	LO4	Hands-on lab; Learner Guide; lab folder Lab15-trading-costs	A7, A8, K7
13:55 - 14:05	T4.3 Forward Testing with Automated Paper Trading The bot loop, market clock, risk checks, order status and logging.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A7, A8
14:05 - 14:30	Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading Build pipeline/risk.py and pipeline/runner.py: `python -m pipeline run --once | --loop` with the market clock, risk checks, kill switch and trade log; run a 2025 forward test; run the strategy live on the paper account for a few cycles; and schedule it with cron, launchd or Windows Task Scheduler.	LO4	Hands-on lab; Learner Guide; lab folder Lab16-paper-trading-bot	A7, A8, K7
14:30 - 14:50	Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution Build pipeline/webhook.py: a webhook receiver that turns TradingView alert JSON into a pipeline decision (validation -> signal -> risk -> execution), test it with a local alert simulator and a self-test, and configure a real TradingView strategy alert.	LO4	Hands-on lab; Learner Guide; lab folder Lab17-tradingview-webhook	A7, K5, K7
14:50 - 15:00	T4.4 Potential Reasons for Unintended Outcomes Overfitting, look-ahead and survivorship bias, cost drag, regime change, data and software errors; the diagnosis method.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A9, K8
15:00 - 15:20	Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live Build `python -m pipeline diagnose` to run automated look-ahead (features and execution), cost-drag, overfitting and survivorship checks on any strategy and on the junior quant's exhibit, and write a PASS / WARN / FAIL report with a go-live verdict.	LO4	Hands-on lab; Learner Guide; lab folder Lab18-diagnose-bias	A9, K8
15:20 - 15:35	Tea Break			
15:35 - 15:45	T4.5 Performance Analysis, Insights and Reporting Tearsheets, backtest versus forward test, insights and the recommendation.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A10
15:45 - 16:10	Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report Build `python -m pipeline report` to compare the research backtest with the runner's forward-test logs, run monitoring checks with thresholds, and generate a tearsheet and a markdown report with data-driven insights and a recommendation.	LO4	Hands-on lab; Learner Guide; lab folder Lab19-performance-report	A8, A10
16:10 - 16:30	Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval Build pipeline/agents.py (analyst, researcher, trader and risk agents; rule-based by default, Claude optional) and pipeline/e2e.py (`python -m pipeline e2e`) that runs every stage in order with gates, then evaluate the pipeline and the role of human oversight.	LO4	Hands-on lab; Learner Guide; lab folder Lab20-agentic-capstone	A7, A10, K1
16:30 - 17:15	Written Assessment (Short Answer Questions) 9 open-ended short-answer questions covering K1 to K9.	All LOs	Individual assessment; Assessment papers	K1 - K9

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
17:15 - 18:30	Practical Performance Assessment 4 practical tasks in the Trading Simulator and TradingView (no coding): hypothesis testing, technical indicators and backtesting, ML model and A/B testing, and the multi-agent desk with forward test, diagnosis and report - covering A1 to A10.	All LOs	Individual assessment; Assessment papers	A1 - A10

Total instructional time, Day 2: 8 hours (480 minutes) = 6 hours training + 2 hours assessment, excluding the tea breaks and the 1-hour lunch break.

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7. Assessment Summary

Participants must be assessed as Competent (C) in BOTH assessment components to be awarded competency for this module. Each component is marked out of 100 against the marking points in the Assessment Rubric; Competent requires at least 80 marks in each component, at least 60% of the marks for every knowledge and ability statement, and every critical marking point awarded.

Method	Duration	Format	TSC Ref	Coverage
Written Assessment (Short Answer Questions)	45 minutes	9 open-ended short answer questions	K1 - K9	All 9 knowledge statements
Practical Performance	75 minutes	4 practical tasks (the Trading Simulator and TradingView, as in the labs - no coding; the Trading Simulator's paper broker or an Alpaca paper account) on a supplied dataset	A1 - A10	All 10 ability statements

Assessment format: Open book. Candidates may refer to their Learner Guide, the lab code and their own class notes. Refer to the Assessment Plan for the full assessment specification, marking guides and records.

8. Training Resources

Trainer Resources

- Trainer slide deck (PowerPoint), 204 slides covering all four topics and twenty labs.
- Lesson Plan (this document) and Facilitator Guide.
- Assessment Plan, including question papers, marking guides, rubric and Assessor Key.
- Lab folder (labs) with all scripts, Pine Script files, bundled market data and reference outputs; a trainer Alpaca paper account for live demonstrations.
- Laptop with Python 3.10+, a modern web browser, projector and whiteboard.

Learner Resources

- Learner Guide containing all topic content and step-by-step instructions for every lab.
- One laptop per participant with Python 3.10+, a modern web browser and internet access; a free Alpaca paper-trading account and a free TradingView account (created in Lab 1 and Lab 2).
- The lab folder, including requirements.txt, the common library and bundled daily market data so every lab also runs offline.
- The course Trading Simulator (https://github.com/alfredang/algorithmic-trader-bot): a local web app (uv or Docker Compose) with a Course Labs page that walks through all twenty labs; the Python and Pine Script files in the lab folders are the Full Code steps.

9. Trainer Notes

- Paper trading only. Never demonstrate with a live-money account, and remind participants that nothing in the course is investment advice.

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- If the venue network blocks the Alpaca API, or a participant cannot open an account, every lab runs offline on the bundled data and the built-in paper-broker simulator. Show both paths early (Lab 1) so nobody is blocked.
- Full Code edition: every lab is built and run in Python from the labs_fullcode folder, with an AI coding assistant and the lab's prompts (LabNN_Prompts.pdf); each lab folder holds the runnable script and the complete reference code. The Trading Simulator remains available for live demonstrations. Have the simulator installed and seeded on the trainer machine before Day 1 so you can project each lab's page.
- Webhook alerts on TradingView need a paid plan. In Lab 17 demonstrate with the trainer's account if available, and let participants use send_test_alert.py, which posts the identical JSON a TradingView alert sends.
- Lab 18 contains deliberately seeded faults. Do not reveal them; let participants find the look-ahead first, because it produces no error message - only a result that is too good.
- Keep the finance context foremost. When a participant reports a great backtest, ask the three questions: is it net of costs, is it out of sample, and what would the investment committee need to see before risking capital?
- In Topic 1, link every milestone of the AI and algorithmic-trading histories to a control used later in the course (circuit breakers to the kill switch, Knight Capital to deployment testing, MiFID II to algorithm records).

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